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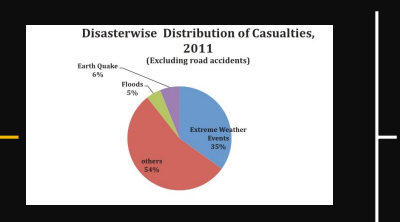
        - 

          common
          - India diaster report, 2011
    - Body
      - diagram
      - causes
      - impacts
        - social
          - High mortality (e.g., Malin 2014, Kedarnath 2013); destruction of settlements and heritage structures
          - Disruption of critical infrastructure (roads, bridges, power lines)
            - the 2021 Chamoli disaster damaged two hydropower plants causing over $200 million in losses
        - economic
          - Disruption of tourism and pilgrimage routes (e.g., Char Dham)
          - Loss of life leading to loss of productivity; massive compensation liabilities; damage to plant property and raw materials; disruption of supply chains
        - envtl
          - loss of biodiversity
          - Contamination of air, soil, and water bodies (surface and groundwater); adverse impact on flora and fauna; long-term ecological degradation
      - challenges
      - Measure
        - adaptation
          - Hazard Zonation: Mapping of GLOF Hazard Zones and restricting settlements/infrastructure in high-risk areas (Land Use Planning)
          - Early Warning Systems (EWS):
        - mitigation
      - Govt programme
        - National level
          - National Policy on Disaster Management;
        - International level
    - conclusion
      - forward looking
        - Technological Integration: Utilizing Synthetic-Aperture Radar (SAR) satellite imagery for all-weather monitoring and AI to detect small changes in lake areas
        - Transboundary Collaboration: Strengthening data-sharing mechanisms with upstream countries (China, Nepal) regarding river levels and glacial lake status
        - Community Focus: Strengthening Community-Based Disaster Preparedness (CBDP) to empower locals as first responders
      - alining with gloabal sdg or framework or programme
        - Align with SDG 13 (Climate Action) by strengthening resilience to climate-related hazards and SDG 11 (Sustainable Cities and Communities) by reducing the number of deaths and economic losses caused by disasters
  - Parlance
    - moraine-dammed glacial lakes
- Diaster Management
  - Old notes
  - Main Topic 14
  - EQ
  - Main Topic 6
  - Drought
  - Main Topic 8
- Landslide
  - Intro
    - Definition by instn.
      - A landslide is defined as "a movement of mass of rock, earth or debris down a slope" under the influence of gravity
      - Indian Institute of Technology Delhi has made the first high resolution landslide susceptibility map for India and the map data is available for free
      - Most common diaster in Himalayas , Western Ghats region causing human, infrastructural and ecological loss
    - types
      - rock falls, topples, debris flows, and creep
    - Data
      - recent event
        - Wayanad Landslides, Kerala (30 July 2024).
        - Sikkim GLOF/Landslide Cascade (October 2023/2024 context).
        - Chamoli
          - flash flood of 2021 in the Chamoli district of Uttarakhand which claimed at least 72 lives was due to an avalanche
        - Himachal
        - Shirur
      - % under prone
        - % Area Prone: Approximately 0.42 million sq. km, covering nearly 12.6% of India's land area (excluding snow-covered area), is prone to landslide hazards
          - Northwest Himalayas: Contributes 66.5% of landslides in India (J&K, Himachal Pradesh, Uttarakhand)
          - Northeast Himalayas: Accounts for 18.8% (Sikkim, Darjeeling, Seven Sisters)
          - Western Ghats: Accounts for 14.7% (Kerala, Karnataka, Goa, Maharashtra)
          - ISRO's Landslide Atlas of India Findings:
  - Body
    - diagram
    - causes
      - Rainfall (Natural): Rainfall is the primary natural triggering factor; variability in rainfall patterns and pore water pressure saturation destabilize slopes
      - Gravity: Gravity is the primary reason for the landslides. Steeper slopes coupled with gravitational force can trigger a massive landslide
      - Seismicity (Natural): Earthquake-induced landslides (co-seismic) occur due to ground shaking which activates existing slides or creates new ones (e.g., Sikkim 2011, Kashmir 2005)
      - Unplanned urbanization/encroachments, deforestation (removing root cohesion), road cutting, and blocking of natural drainage patterns significantly increase vulnerability
      - Natural
        - Earthquake , GLOF , fragile ecology , rainfall
      - Manmade
        - mining and quarrying
          - Vibrations from the blasts can degrade soils in the locations prone to landslides, such as in the Himalayas.
            - Idduki and Waynad
        - unplanned hill cutting
          - Chardham
        - lack of Slope stability checks
          - Mid slope hotels and hydropower dams
        - Urban encroachment into forests, disturbance of natural drainage, and cutting and deep excavations on hill slopes to develop buildings and infrastructure have contributed to a huge increase in landslides in Uttarakhand
          - Joshimath in 2023 was mainly due to a lack of infrastructure for drainage, an increase in human population, and rapid expansion of infrastructure to cater for the tourism potential.
        - Climate change: The intense and sudden rainfall due to climate change as well as the melting of the glaciers in the Himalayas have aggravated the problems of landslides in India
    - impacts
      - social
        - Global Rank: India is among the top 5 countries with the highest landslide risk, where the estimated loss of life per 100 sq. km per year is greater than one
          - ISRO Landslide Atlas of India
        - High mortality (e.g., Malin 2014, Kedarnath 2013); destruction of settlements and heritage structures
        - Disruption of critical infrastructure (roads, bridges, power lines)
          - the 2021 Chamoli disaster damaged two hydropower plants causing over $200 million in losses
      - economic
        - Globally, landslides contribute to an estimated average annual economic loss of $26 billion
        - Disruption of tourism and pilgrimage routes (e.g., Char Dham)
      - environmental
        - Formation of Landslide Dams (e.g., Parechhu river blockage) which pose flash flood risks upon breaching,. Loss of topsoil and biodiversity
    - challenges
      - Technical: Lack of high-resolution (1:10,000 scale) hazard maps for local planning
      - Difficulty in real-time monitoring and forecasting due to the localized nature of landslides
      - Administrative: Implementing and enforcing building codes and land-use regulations in difficult mountainous terrain
    - Measure
      - adaptation
        - Structural
          - Slope Stabilization: Use of retaining walls, soil nailing, rock bolting, and anchors
          - Drainage Correction: Construction of catch-water drains, contour drains, and sub-surface drainage to manage pore water pressure
          - Bio-engineering: Afforestation and use of geosynthetics (e.g., coir logs, jute) to arrest soil erosion Landslide Monitoring involves deploying sensors, satellite imagery, and ground-based instruments to detect ground movement and potential landslide precursors.
          - Landslide Monitoring involves deploying sensors, satellite imagery, and ground-based instruments to detect ground movement and potential landslide precursors.
            - Landslip Detection System: It was developed by the Indian Institute Of Technology-Mandi (IIT-Mandi), the Indian Army, and the Defence Research and Development Organisation (DRDO)
        - Non-structural
          - Land Use Zoning: Restricting development in high-hazard zones and strictly enforcing building codes
          - Early Warning Systems (LEWS): Developing rainfall threshold-based warning systems (e.g., I-D models)
      - mitigation
    - Govt programme
      - National level
        - National Landslide Risk Management Strategy (NLRMS) 2019: Addresses hazard mapping, monitoring, and capacity building
          - The NLRMS aligns with the Sendai Framework to enhance resilience, but effective implementation requires interstate coordination and local participation.
        - National Landslide Susceptibility Mapping (NLSM): Programme by Geological Survey of India (GSI) to map landslide susceptibility on a 1:50,000 scale
        - Landslide Atlas of India (2023): Released by ISRO/NRSC, providing a database of ~80,000 landslides for risk analysis
        - NDMA Guidelines (2009): Framework for management of landslides and snow avalanches
      - International level
        - Sendai Framework: Focuses on understanding disaster risk and investing in DRR for resilience
        - Coalition for Disaster Resilient Infrastructure (CDRI): Promotes resilient infrastructure in mountain regions
  - conclusion
    - forward looking
      - Development of mobile apps like 'Bhuvan FLIM' for field data collection
      - Training local communities (Naula/Dhara maintenance) and integrating indigenous knowledge for early warning and response
      - Generation of User-Friendly Landslide Hazard Maps
        - Utilizing InSAR (Interferometric Synthetic Aperture Radar), LiDAR, and UAVs (Drones) for precise slope monitoring and kinematics analysis
      - Following International Best Practices: Brazil's SNAKE System: The SNAKE System is the Landslide Early Warning System (LEWS),
    - alining with gloabal sdg or framework or programme
      - Aligns with SDG 11 (Sustainable Cities and Communities) by reducing the number of deaths and economic losses caused by disasters, and SDG 15 (Life on Land) by restoration of degraded land and soil
    - Conclusion
      - Landslides are not just geological, but governance challenges — integrating science, community, and policy is the only sustainable path.
      - Mountain Criers Before it falls
- Main Topic 11
- GLOF
  - Intro
    - Definition by instn.
      - GLOFs occur when a substantial amount of water stored in a glacial lake is suddenly discharged.
      - It involves the release of millions of cubic metres of water and debris in a short period, causing catastrophic flooding downstream
    - Data
      - recent event
        - The recent breaking off of a portion of the Nanda Devi glacier in Joshimath, Uttarakhand, resulted in significant flooding in parts of the state.
        - Event: Sikkim GLOF (October 2023 - Context for current mitigation).
          - Triggered by South Lhonak Lake burst; destroyed 31 major bridges, damaged the Chungthang Dam
        - Global Context: 2025 has been designated as the "International Year of Glaciers' Preservation" by the UN to address cryosphere risks
      - % under prone
        - Globally, 15 million people are exposed to impacts from potential GLOFs, with the highest exposure in the High Mountain Asia region
  - Body
    - diagram
    - causes
      - Fast Slope Migration: Rapid slope movement (slides, falls, avalanches) into the lake leads to displacement waves, causing the dam to overtop or triggering an immediate dam breach
        - For instance, 50% of GLOFs in the Hindu Kush Himalaya are triggered by ice avalanches
          - NDMA
      - Heavy Rainfall/Snowmelt and Cascading Processes: Intense rainfall or rapid snowmelt, along with cascading processes, result in a significant increase in water intake into the lake, potentially leading to a flood from an upstream lake.
      - Earthquake: Earthquakes can directly cause dam breaches and failures, resulting in a GLOF
      - Climate change
        - retreat of glaciers and formation of pro-glacial lakes
        - permafrost thaw
        - Black Carbon deposition reduces albedo, accelerating melt
    - impacts
      - social
        - Catastrophic loss of life and displacement in downstream communities. The events are sudden, often giving little lead time for evacuation
      - economic
        - GLOFs carry massive sediment loads that silt up reservoirs, reducing their economic life
          - Severe damage to Hydro-electric Projects (HEPs) (e.g., Chamoli 2021, Sikkim 2023)
      - environmental
        - Extreme channel erosion, sedimentation of riverbeds, and destruction of forests and arable land downstream
    - challenges
      - Monitoring Constraints: Inaccessibility of high-altitude terrain makes ground-based monitoring difficult; reliance on remote sensing often faces issues with cloud cover during monsoons
      - Short Lead Time: GLOF events can occur within minutes, challenging standard early warning dissemination protocols
    - Measure
      - adaptation
        - Lake Level Lowering: Pumping or siphoning water out of the lake to reduce hydrostatic pressure. Controlled breaching of moraine dams
        - Channel Construction: Construction of spillways or outlets to regulate water discharge (e.g., work done at South Lhonak Lake)
        - Check Dams: Construction of check dams and deflection walls downstream to arrest debris flow and reduce impact energy
        - Early Warning Systems (EWS): Installation of Automatic Water Level Recorders (AWLR) and sensor-based alerts
      - mitigation
    - Govt programme
      - National level
        - National GLOF Risk Mitigation Programme: Led by NDMA, this focuses on the risk inventory of 195 lakes, installation of automated weather stations, and remote sensing
        - NDMA Guidelines (2020): Comprehensive guidelines on the Management of Glacial Lake Outburst Floods, emphasizing hazard assessment and monitoring
        - Hydro-Meteorological Action Plan (HMAP): J&K initiative involving AWLRs and Automatic Rain Gauges for real-time data
      - International level
  - conclusion
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- Main Topic 13
- industrial diaster
  - Intro
    - Definition by instn.
      - Chemical disasters are scenarios involving disease, disability, or death on a large scale among humans, animals, and plants due to toxins or disease caused by chemical agents. They may result from fire, explosion, toxic release, or poisoning
      - Genesis Event: The Bhopal Gas Disaster (December 1984), where the release of Methyl Isocyanate killed over 2,000 people immediately and affected thousands long-term, brought a paradigm shift in Chemical Disaster Management (CDM) in India
    - Data
      - recent event
        - HPCL Refinery, Vishakhapatnam (1997): Vapour cloud explosion
        - IPCL, Nagothane (1990): Gas Cracker Complex explosion
      - % under prone
        - MAH Units: There are approximately 1,666 Major Accident Hazard (MAH) units in India
  - Body
    - diagram
    - causes
      - These MAH units handle a wide range of chemicals (raw materials, products, wastes) with flammable, explosive, corrosive, and toxic properties
      - Large quantities of chemicals are stored or processed in industries often located in densely populated areas, enhancing vulnerability due to haphazard construction
      - Process and Safety System Failures: Including technical errors (design defects, corrosion, metal failure), human errors (neglecting safety instructions), and organizational errors (poor emergency planning)
      - Natural Calamities: Natural hazards triggering chemical accidents (Natech), e.g., the release of acrylonitrile at Kandla Port during the 2001 earthquake or damage to phosphoric acid containment during the 1999 Orissa Super Cyclone
      - Transportation Accidents: Incidents during the transport of hazardous chemicals (HAZCHEM) by road, rail, air, or pipelines (e.g., BLEVE - Boiling Liquid Expanding Vapour Explosion)
    - impacts
      - social
        - Large-scale mortality and morbidity; long-term health effects such as genetic disorders, neurological deficiencies, and psychological trauma (Post-Traumatic Stress Disorder)
      - economic
        - Loss of life leading to loss of productivity; massive compensation liabilities; damage to plant property and raw materials; disruption of supply chains
      - environmental
        - Contamination of air, soil, and water bodies (surface and groundwater); adverse impact on flora and fauna; long-term ecological degradation
    - challenges
      - Lack of emphasis on CDM (chemical diaster managemnet) functions; deficiencies in On-Site and Off-Site emergency plans; inadequate enforcement of regulations
      - Technical: Absence of national-level risk assessment criteria; use of obsolete technologies by Small and Medium Enterprises (SMEs); lack of standardized safety audit procedures
      - Legal: Lack of legislation on land-use policy (buffer zones) around industries; absence of national regulations specifically for medical emergency management in chemical disasters
      - Infrastructural: Inadequate medical infrastructure for handling mass chemical casualties (shortage of specific antidotes, trained paramedics)
    - Measure
      - adaptation
        - Land Use Planning: Strict implementation of land-use policies to prevent settlements near hazardous industries.
        - 2. Risk Assessment: Mandatory Hazard Analysis (HAZAN) and Hazard and Operability Studies (HAZOP) for identification of risks.
        - 3. Awareness Generation: Sensitizing the community and stakeholders about "DOs and DON'Ts" during chemical accidents.
        - 4. Capacity Building: Training of personnel, including drivers of HAZMAT vehicles, and strengthening of the factory inspectorates
        - Tech Integration: Implementation of GIS-based Emergency Planning and Response Systems in major industrial clusters
        - Development of a National Register on potentially toxic chemicals
        - Establishing a National Action Plan on Chemical Industrial Disaster Management (NAP-CIDM) and ensuring rigorous
      - mitigation
    - Govt programme
      - National level
        - Legal Framework: Manufacture, Storage and Import of Hazardous Chemicals (MSIHC) Rules, 1989
        - Crisis Groups: Establishment of Central (CCG), State (SCG), District (DCG), and Local Crisis Groups (LCG) for coordination
        - Schemes: "Industrial Pocket-wise Hazard Analysis" by MoEF to identify accident potential in districts
        - Institutions: Setting up of Emergency Response Centres (ERCs) and Poison Control Centres
      - International level
        - ILO Convention: Adherence to ILO Convention No. C 174 on prevention of major industrial accidents
  - conclusion
    - forward looking
    - alining with gloabal sdg or framework or programme
- To Do
  - Sendai
- Sendai
  - Intro
    - The Sendai Framework is a global, voluntary agreement adopted in 2015 to substantially reduce disaster risk and losses
      - by shifting focus from disaster response to prevention, preparedness, and resilience.
  - Body
    - what is it
      - Adopted at the Third UN World Conference on Disaster Risk Reduction (2015, Sendai, Japan).
      - Successor to the Hyogo Framework for Action (2005–2015).
      - Led by UN Office for Disaster Risk Reduction (UNDRR).
      - Timeframe: 2015–2030.
    - objective
      - Prevent new disaster risks, reduce existing risks, and strengthen resilience.
    - 4 priorities
      - Understanding disaster risk (data, risk mapping).
      - Strengthening disaster risk governance.
      - Investing in disaster risk reduction (structural & non-structural).
      - Enhancing disaster preparedness for effective response & "Build Back Better".
    - 7 targets
      - Reduce disaster mortality, affected people, economic loss, and damage to critical infrastructure.
      - Increase DRR strategies, international cooperation, and early warning coverage.
    - India and Sendai
      - ~75% of India's districts are vulnerable to extreme events (floods, droughts, cyclones, earthquakes).
      - India is among the top 10 disaster-prone countries globally.
      - India loses ~2–3% of GDP annually due to disasters (World Bank estimates).
    - Issues and challenges in implementation
      - A. High Exposure Due to Geography & Demography
        - Long coastline, Himalayan belt, floodplains.
      - B. Rapid Urbanisation & Unplanned Growth
        - Expansion into floodplains, wetlands, seismic zones.
        - Example: Urban flooding in Chennai (2015) and Bengaluru due to poor planning.
      - C. Weak Local-Level Capacity
        - Disaster management plans exist but poorly implemented at district/ULB level.
      - D. Inadequate Risk-Informed Development
        - Infrastructure projects often ignore hazard mapping.
          - Example: Hydropower & road projects in Himalayas increasing landslide risks.
      - E. Climate Change Multiplying Risks
        - Data: Extreme rainfall events increased by ~75% since 1950.
    - impacts apart from usual ones
      - C. Poverty & Inequality
        - Disasters push vulnerable populations into poverty traps.
      - D. Disaster–Development Vicious Cycle
        - Rebuilding diverts funds from development. leading to slow porgress on sdg
      - Economic Risk: By 2100, India is projected to lose around 3-10% of its GDP annually due to climate change impacts
      - Sea Level Rise: Millions in coastal cities like Mumbai, Kolkata, and Chennai face risks from sea-level rise, flooding, and salinization
        - 3.4 millimeters (0.13 inches) per year in recent decades (1993-2024),
          - NASA
    - India preparedness
      - NDMA , 2005
        - Features
          - National Diaster Response Fund NDRF
            - 100:0 (center state share)
              - supplements diaster relief when diaster is classified as severe
          - SDRF
            - Hilly and NE
              - 90:10
            - other
              - 75:25
            - usage
              - immediate relief for food, shelter, medical care and compensation
          - Funds
            - issues
              - delayed and underapproved funds under NDRF
                - kerala demand
                  - 2200 Cr
                - kerala recieved in waynad
                  - 260 Cr
              - SDRF
                - funds usage restricted to immediate relief but not reconstruction or livlihood restoration
              - Outdated compensation
                - life lost
                  - 4L
                - fully damaged house
                  - 1.2L
              - discretion of center in declaring diaster as severe in absense of definition
              - aid release is procedural not automatic
              - weak allocation crieteria of FC ignoring local vulnerability and adaptability , exposure, etc.
          - Solution
            - Phillipines model of quick response funds through rainfall and fatality indices
            - FC
              - update relief norm cost,
              - revise allocation
                - comprehensive vulnerability index
              - disater assistance remaind grant based not debt based
            - procedural chairty to rules based partnership
          - NDResponse forces, SDRF
          - National Disaster Management Authority (NDMA)
            - → Policy formulation, guidelines aligned with Sendai priorities.
      - National Disaster Management Plan (2016, updated)
        - → India's roadmap aligned with Sendai Framework.
      - Early Warning Systems
        - → IMD-led cyclone & weather warnings; success in Odisha cyclone evacuations.
      - Coalition for Disaster Resilient Infrastructure (CDRI)
      - National Cyclone Risk Mitigation Project (NCRMP)
        - → Strengthens coastal resilience.
    - way ahead
      - Mainstream DRR into all development planning.
      - Strengthen urban resilience & risk-sensitive land-use planning.
      - Invest in local-level capacity & community preparedness.
      - Expand multi-hazard early warning coverage to last-mile.
      - Integrate climate adaptation with DRR.
      - Improve data sharing, risk modelling, and monitoring.
  - Conclusion
    - Effective implementation of the Sendai Framework through risk-informed development, strong local institutions, and climate-resilient planning is crucial to break India's disaster–development cycle and safeguard long-term growth.
    - It aligns closely with the Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities and Communities), aiming to substantially reduce disaster risk and losses in lives, livelihoods, and health

Presented with xmind